



Suricata Cheat Sheet

A quick-reference guide for IDS/IPS rule syntax, flow matching, protocol inspection, logging, and alerting concepts in Suricata.

Rule Structure

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PART	MEANING
action	Alert, pass, drop, reject, accept, etc.
proto	Base protocol: tcp, udp, icmp, ip (http, tls, dns are app-layer over these).
src / dst	Source and destination addresses or networks (e.g., \$HOME_NET, \$EXTERNAL_NET).
port	Source and destination service ports.
direction	->, <-, or <>.
options	Matching logic, metadata, payload content, flow state.

Core Actions

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ACTION	BEHAVIOR
alert	Generate an alert, continue processing.
drop	Block packet and log it (IPS mode).
reject	Drop and send TCP reset / ICMP unreachable.
pass	Bypass further detection for this packet/flow.
accept	Let traffic pass; used in firewall/IPS hook rules.
sdrop	Silently drop without logging.

Common Rule Keywords

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KEYWORD	PURPOSE
msg	Human-readable alert message.
sid	Signature ID; unique rule identifier.
rev	Revision number for the rule.
classtype	Category such as trojan-activity or web-application-attack.
reference	External intel sources: cve, url, bugtraq, nessus, etc.
metadata	Extra tagging or context (e.g., created_at, former_category).

Match Conditions

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CONDITION	EXAMPLE
content	content:"GET";
pcre	pcre:"/evil/i"; (supports modifiers like /i)
byte_test	Check numeric field at offset/length.
isdataat	Verify data exists at relative offset.
depth	Limit search to first N bytes of payload.
distance	Search relative to a previous content match.

Flow, Session & Related

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KEYWORD	MEANING
flow:	to_server, to_client, established, stateless.
ip_proto	Match on IP protocol number.
ttl	Check packet TTL value.
sameip	Require same source and destination IP.
stream	TCP stream reassembly for content matching.
app-layer	Protocol-aware parsing (HTTP, DNS, TLS, SMB, etc.).

Protocol Examples

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PROTOCOL	TYPICAL USE
tcp	Application and port-based rules.
udp	DNS, NTP, SNMP, DHCP, VOIP checks.
icmp	Echo, unreachable, scan or reconnaissance detection.
http	Request/response headers and URIs (app-layer over tcp).
tls	SSL/TLS handshake and certificate inspection (app-layer).
dns	Domain and record-based detections (app-layer).

Alert & Log Controls

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KEYWORD	ROLE
threshold	Rate-limit repeated alerts (type: limit/threshold/both; track: by_src/by_dst).
metadata	Tag severity, threat, or source context.
flowbits	Track state across multiple packets per flow.
noalert	Match without generating an alert.
tag	Add metadata or tracking to alert events.
priority	Set importance of the rule.

Example Rule

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COMPONENT	EXAMPLE
Header	alert http \$HOME_NET any -> \$EXTERNAL_NET any
Flow	flow:to_server,established;
Buffer + Match	http.uri; content:"/update.bin";
Metadata	classtype:trojan-activity;
IDs	sid:1000001; rev:1;

Useful Rule Examples

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SCENARIO	EXAMPLE
HTTP GET keyword	alert http any any -> any any (msg:"HTTP GET observed"; http.method; content:"GET"; sid:1000001; rev:1;)
Suspicious PowerShell	alert tcp any any -> any any (msg:"PowerShell command"; flow:to_server,established; content:"powershell"; nocase; sid:1000002; rev:1;)
Known malicious IP	alert ip 203.0.113.66 any -> any any (msg:"Known malicious IP"; sid:1000003; rev:1;)
TCP port scan guard	alert tcp any any -> any any (msg:"TCP port scan"; flags:S; flow:stateless; threshold:type both, track by_src, count 5, seconds 30; sid:1000004; rev:1;)
Malicious DNS query	alert dns any any -> any any (msg:"Known malicious domain"; dns.query; content:"evil-domain.com"; nocase; sid:1000005; rev:1;)
Suspicious TLS SNI	alert tls any any -> any any (msg:"Suspicious TLS SNI"; tls.sni; content:"malware-c2.net"; nocase; sid:1000006; rev:1;)
SQL injection in URI (pcre)	alert http any any -> any any (msg:"Possible SQL injection"; flow:to_server,established; http.uri; pcre:"/(union[[:space:]]+select '[[[:space:]]+or[[[:space:]]+1=1)/i"; sid:1000007; rev:1;)
SQLmap user-agent	alert http any any -> any any (msg:"SQLmap user-agent"; flow:to_server,established; http.user_agent; content:"sqlmap"; nocase; sid:1000008; rev:1;)
XSS in URI	alert http any any -> any any (msg:"Executable downloaded over HTTP"; flow:to_client,established; fileext:"exe"; filestore; sid:1000010; rev:1;)
Executable download (file magic / store)	alert http any any -> any any (msg:"Executable downloaded over HTTP"; flow:to_client,established; fileext:"exe"; filestore; sid:1000010; rev:1;)
SSH brute force (threshold)	alert tcp any any -> any 22 (msg:"Possible SSH brute force"; flow:to_server,established; content:"SSH-"; threshold:type both, track by_src, count 15, seconds 60; sid:1000011; rev:1;)
FTP login failures	alert tcp any any -> any 21 (msg:"Possible FTP brute force"; flow:to_client,established; content:"530 "; nocase; threshold:type both, track by_src, count 10, seconds 60; sid:1000012; rev:1;)
ICMP ping sweep	alert icmp any any -> any any (msg:"ICMP echo request (ping)"; itype:8; sid:1000013; rev:1;)
SYN-only recon / scan (stateless)	alert tcp any any -> any any (msg:"SYN-only packet (recon/scan)"; flags:S; flow:stateless; threshold:type both, track by_src, count 20, seconds 10; sid:1000014; rev:1;)
SMB named pipe (depth)	alert tcp any any -> any 445 (msg:"SMB named pipe request"; flow:to_server,established; content:" ff 53 4d 42 "; depth:4; sid:1000015; rev:1;)
Correlated two-stage detection (flowbits)	alert tcp any any -> any any (msg:"Stage 1 - suspicious download"; flow:to_server,established; content:"/update.bin"; flowbits:set,stage1_dl; flowbits:noalert; sid:1000016; rev:1;) alert http any any -> any any (msg:"Stage 2 - callback detected"; flow:to_client,established,http.status_code; content:"200"; flowbits:isset,stage1_dl;sid:1000017;rev:2;)
Modbus (ICS / OT) register read	alert modbus any any -> any any (msg:"Modbus read holding registers"; modbus.function; content:" 03 "; sid:1000018; rev:1;)
Full-featured rule (metadata / classtype)	alert tcp \$HOME_NET any -> \$EXTERNAL_NET any (msg:"ET TROJAN Possible downloader"; flow:established,to_server; content:" 90 90 90 "; depth:6; classtype:trojan-activity; reference:url,example.com/analysis; metadata:created_at 2026_01_01, former_category MALWARE; priority:1; sid:1000019; rev:2;)